

The logic of arguments - Set 1 - Solutions

Specification Arg1-Arg4.

Answer key

Question	Answer	Topic
1	A	The logic of arguments
2	C	The logic of arguments
3	D	The logic of arguments
4	C	The logic of arguments
5	C	The logic of arguments
6	C	The logic of arguments
7	C	The logic of arguments
8	C	The logic of arguments
9	E	The logic of arguments
10	D	The logic of arguments
11	B	The logic of arguments
12	A	The logic of arguments

Worked solutions

Question 1 LGC-01 · The logic of arguments · spec Arg1

Consider the statement: **if n is a multiple of 6, then n is a multiple of 3.** Which of the following is its contrapositive?

Answer: A

The contrapositive of 'if A then B ' is 'if not B then not A '. It is logically equivalent to the original; B is the converse and C is the inverse.

Question 2 LGC-02 · The logic of arguments · spec Arg2

Let P be the statement ' $x > 2$ ' and Q the statement ' $x^2 > 4$ ', where x is a real number. Which one of the following is true?

Answer: C

$x > 2 \Rightarrow x^2 > 4$, so P is sufficient for Q . But $x = -3$ gives $x^2 > 4$ without $x > 2$, so P is not necessary.

Question 3 LGC-03 · The logic of arguments · spec Arg4

Which of the following is the negation of the statement **'every student in the class scored more than 50' **?

Answer: D

The negation of 'for all x , $P(x)$ ' is 'there exists x such that not $P(x)$ '. 'Not more than 50' means '50 or less'.

Question 4 LGC-04 · The logic of arguments · spec Arg1

Consider the statement about a real number x : **if $x^2 = x$, then $x = 1$.** Which one of the following is true?

Answer: C

$x = 0$ satisfies $x^2 = x$ but not $x = 1$, so the statement is false. The converse, 'if $x = 1$ then $x^2 = x$ ', is true.

Question 5 LGC-07 · The logic of arguments · spec Arg2

Let P : 'the quadrilateral $ABCD$ is a square' and Q : 'the quadrilateral $ABCD$ has four equal sides'. Which one of the following is true?

Answer: C

A square has four equal sides ($P \Rightarrow Q$, sufficient), but a rhombus that is not a square shows Q does not imply P (not necessary).

Question 6 LGC-08 · The logic of arguments · spec Arg4

Which of the following is the negation of **'there exists a positive integer n such that $n^2 + n + 41$ is not prime' **?

Answer: C

Negating 'there exists n with not $P(n)$ ' gives 'for all n , $P(n)$ '.

Question 7 LGC-10 · The logic of arguments · spec Arg2

For real numbers a and b , let P : ' $a = b$ ' and Q : ' $a^2 = b^2$ '. Which one of the following is true?

Answer: C

$a = b \Rightarrow a^2 = b^2$ (sufficient), but $a = 1, b = -1$ gives $a^2 = b^2$ without $a = b$ (not necessary).

Question 8 LGC-11 · The logic of arguments · spec Arg1

The statement '**if a triangle is equilateral then it is isosceles**' is true. Which of the following must therefore also be true?

- I If a triangle is isosceles then it is equilateral.
- II If a triangle is not isosceles then it is not equilateral.
- III If a triangle is not equilateral then it is not isosceles.

Answer: C

Only the contrapositive (II) is logically equivalent to the original. I is the converse and III is the inverse; both are in fact false here.

Question 9 LGC-13 · The logic of arguments · spec Arg4

Which of the following is the negation of **' $x > 3$ and $y \leq 5$ '**?

Answer: E

Not (A and B) is (not A) or (not B). Not $x > 3$ is $x \leq 3$; not $y \leq 5$ is $y > 5$.

Question 10 LGC-14 · The logic of arguments · spec Arg1

For a positive integer n , consider the statement: **'if n is prime, then n is odd.'** Which one of the following is true?

Answer: D

$n = 2$ is an even prime, so the statement is false. The converse, 'if n is odd then n is prime', fails for $n = 9$.

Question 11 LGC-17 · The logic of arguments · spec Arg1

Let x be a real number. Which of the following statements is equivalent to **' $x \geq 2$ if and only if $x^3 \geq 8$ '**?

Answer: B

'If and only if' is the conjunction of the implication and its converse.

Question 12 LGC-19 · The logic of arguments · spec Arg2

Let f be a function defined for all real numbers. P : ' $f'(a) = 0$ '; Q : ' f has a stationary point at $x = a$ '. Which one of the following is true?

Answer: A

A stationary point is by definition a point where the derivative is zero, so P and Q are equivalent.